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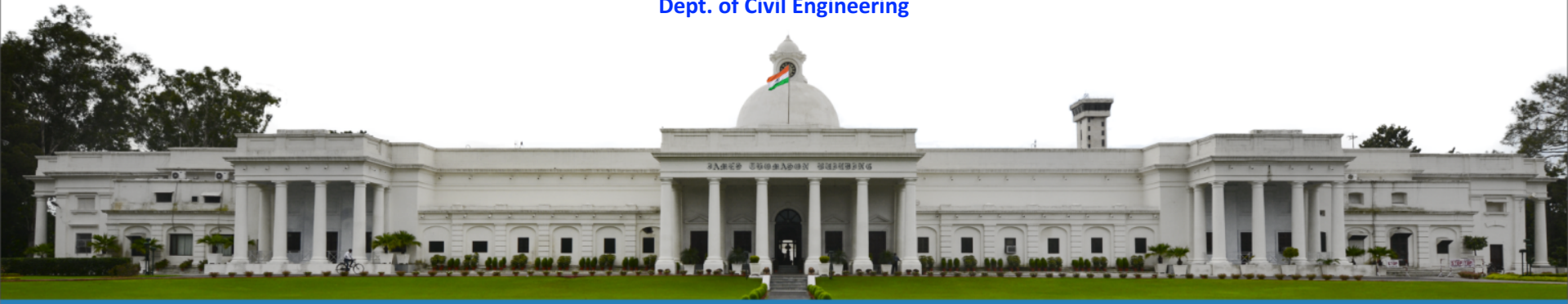
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SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 14: Methodologies of EIA

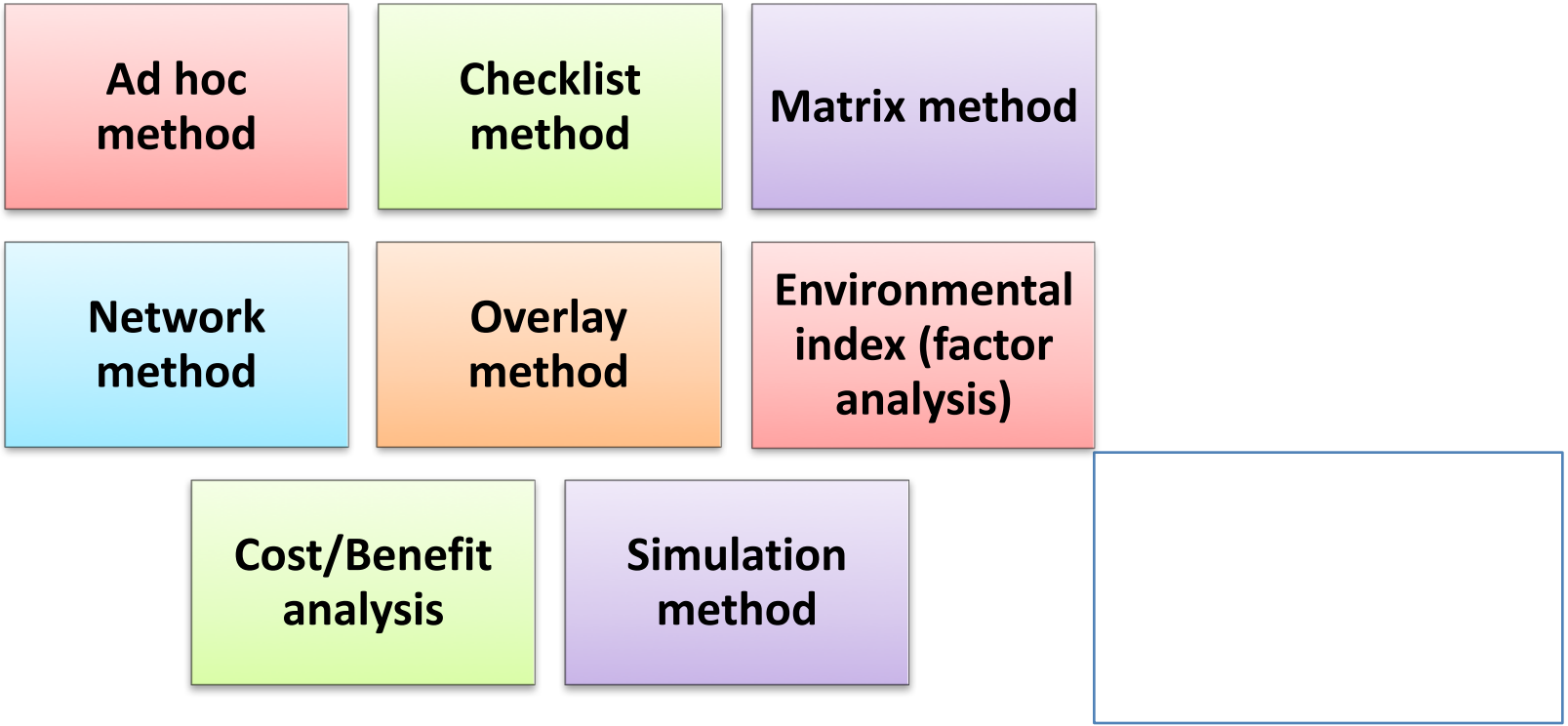
By-
Prof. Bhola Ram Gurjar
Dept. of Civil Engineering



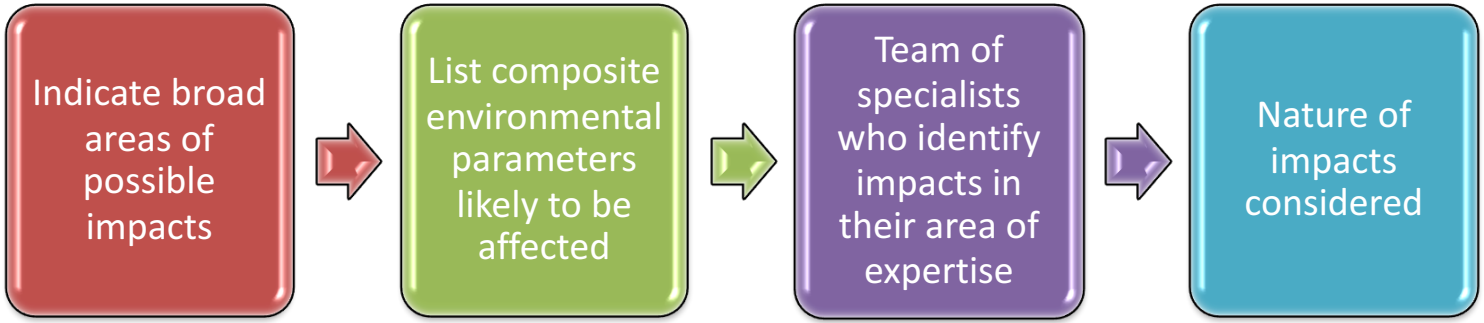
Contents

- Methodologies for Environmental Impact Assessment
  - Ad-hoc method
  - Checklist method
  - Matrix method
  - Network method
  - Overlay method
  - Simulation method
  - Cost-Benefit Analysis method
  - Environment Index method
- Benefits and limitations
- Examples
- Conclusion

Methodologies for Environmental Impact Assessment



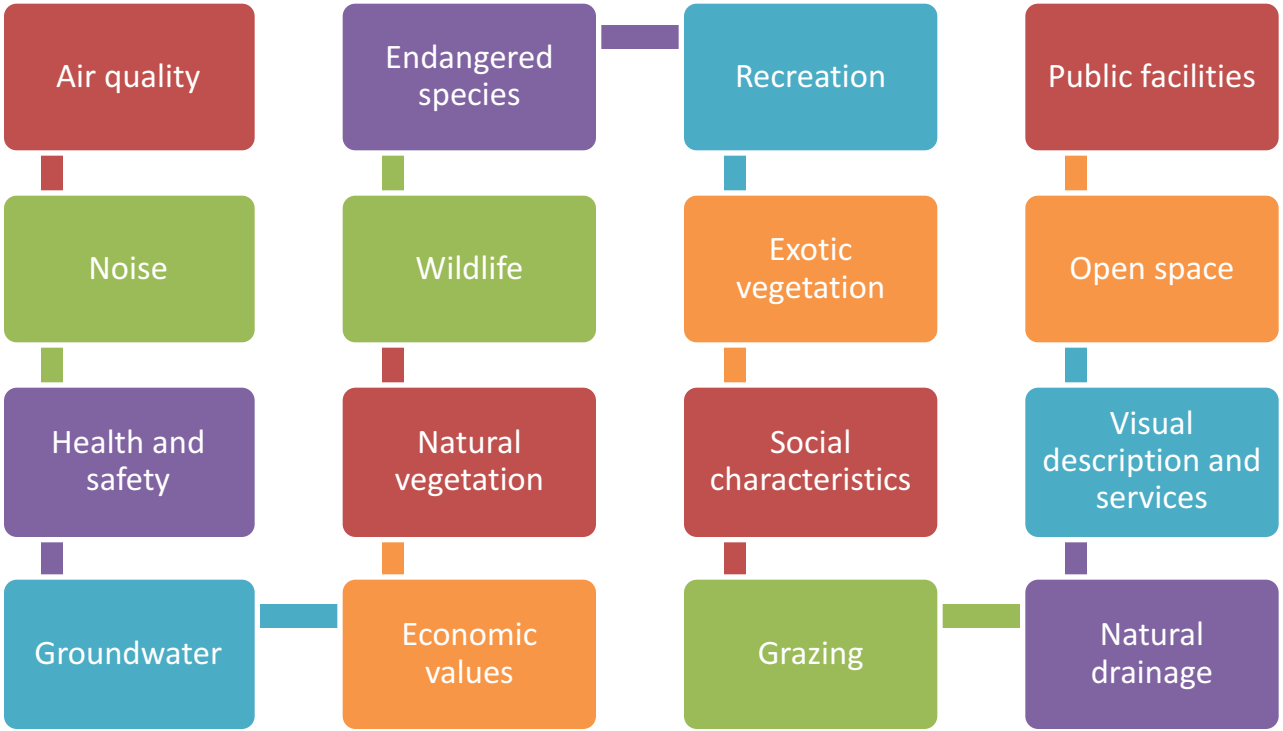
Approach for Ad-hoc Method



Features of Ad-hoc Method

Intuitive& qualitative. Generally used in screening process.

Common Criteria for Consideration

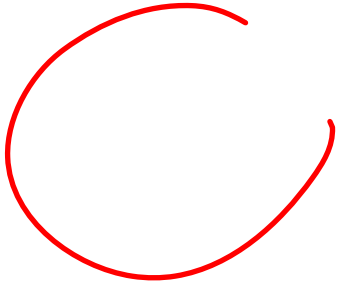


Types of Ad-hoc Methods

Common practice:

Committee of expert decides based on consensus or majority.





Ad-hoc Method Practices

U.S.A.

Assemble a team of specialists to identify impacts in their areas of expertise, with minimal guidance beyond the requirements of the NEPA (National Environmental Protection Act)

India

Similar process conducted during Screening process to decide whether to do Impact Assessment or not.

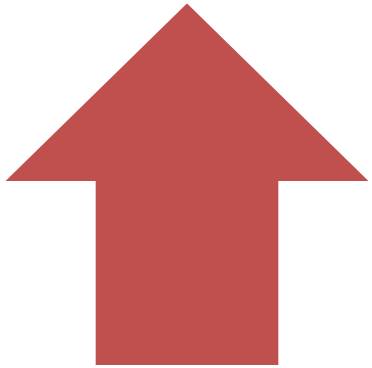
Isn't the example Ad-hoc process? (Yes, It is..)

Alternative Reservoir: Ad-hoc Method

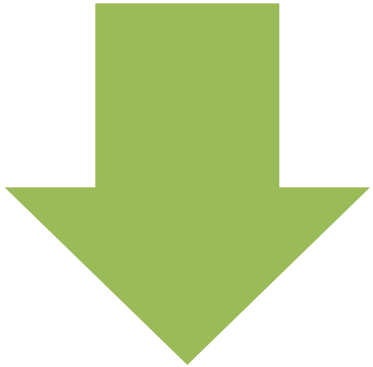
Parameters	Alternative-1	Alternative-2	Alternative-3	Alternative-4
Air Quality				
Noise				
Wildlife				
Local Population				
Water Quality				

Rank 1/2/3/4 to each alternative on every parameters and assess the best option among alternatives

Pros & Cons of Ad-hoc Method



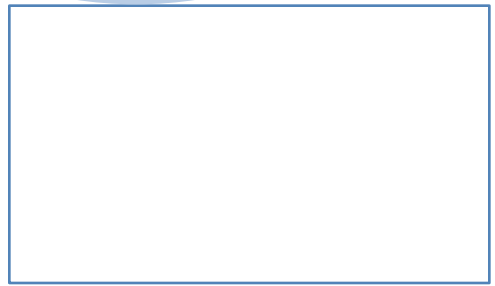
- Simple and can be performed without any training
- No relative weighting or cause-effect relationship
- Suggests only broad areas for possible impacts



- No assurance that all relevant impacts have been studied
- Lacks consistency
- Inefficient as it requires a considerable effort to identify and assemble a panel for each assessment

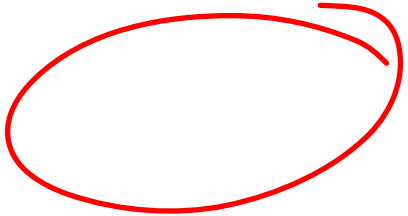
Check-List Method

Mainly used during Screening & Scoping stages of EIA process.



Types of Check-List Method

Checklists are strong indicators of impact identification. They effectively garner the attention and awareness of their audience



Simple Check-List

- List of parameters
- Without guidelines regarding either interpretation
- Further needs measurement of environmental parameters or specific data needs or impact prediction and assessment



Example of Simple Check-List

Environmental elements	Environmental impact factors	Construction period	Operating period
Social environment	Traffic, development of social economy	○	★
	Land use	☆	☆
	Demolition and resettlement, traffic, irrigation works, social economy	★	○
Living environment	Traffic, exchange, employment, income and safety	★	○
Ecological environment	Water and soil loss	★	○
water environment	Water pollution during construction period and operation period, accident risk	☆	○
Acoustic environment	Traffic noise	☆	★
Air quality	Raise dust、TSP	★	○
	Harmful substances in tail gas（NOx、CO）	○	☆

Solid Star: Obvious effects; Star: Usual effects; Circle: Slight effects

Source: world Bank, 2011

Example of a Simple Check-list for a project of Township

Descriptive Check-List

- Include list of environmental factors
- Information on measurement, impact prediction and assessment



Example of Descriptive Check-List

Parameters	Description
Air Quality	Current condition (PM_{10}, $PM_{2.5}$, NO_X, SO_X), area affected, no. of people affected
Water Quality	Current status of water bodies, Dissolved Oxygen level, possible pollutants, Effects on population
Noise	Change in noise level, frequency of occurrence

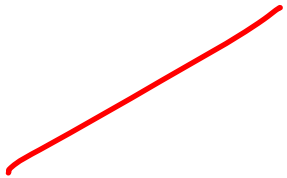
Scaling/Weighting Check-List

- Include list of environmental factors along with importance of factors.
- Weight is given as per importance.
- Require experts for scaling the various environmental factors.



Key Features of Scaling Check-List







Example of Scaling Check-List

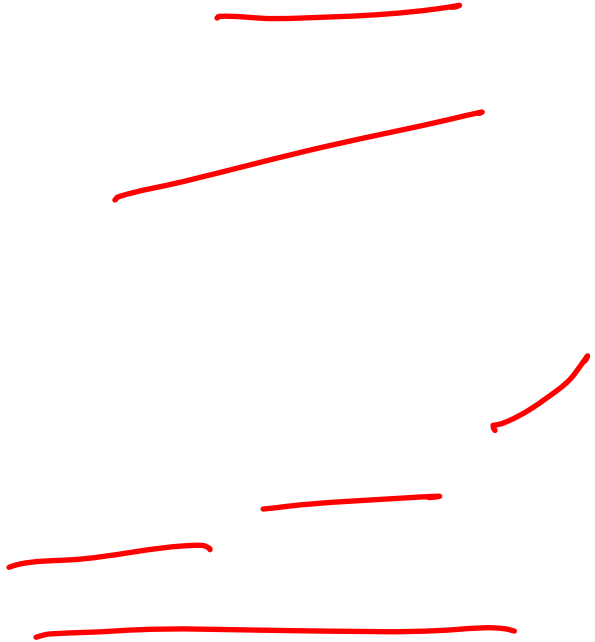
Parameters	Weightage	Alternative-1	Alternative-2
Air Quality	40		
Noise	15		
Wildlife	10		
Local Population	20		
Water Quality	15		
Total	100		

Limitations of Scaling Check-List

- Pose the danger of imparting equal importance to every impact.
- Numerical values assigned to impacts can be derived based only on subjective understanding.
- Make no provision for assessing dynamic probabilistic trends or mitigation, enhancement and monitoring.
- Can't identify higher order effects, impacts and interactions.

Parameters	Weightage	Alternative-1	Alternative-2
Air Quality	40		
Noise	15		
Wildlife	10		
Local Population	20		
Water Quality	15		
Total	100		

Pros & Cons of Check-List Method



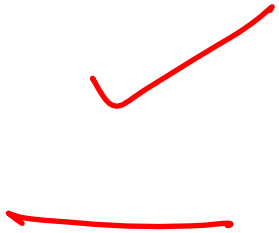
Matrix Method



Example of Matrix Method

Environmental Components	Project Action				
	Construction		Operations		
	Utilities	Buildings (Residential + Commercial)	Residential	Commercial	Parks & Open Space
Flora					
Fauna					
Air Quality					
Water Quality					
Population Density					
Employment					
Traffic					

Pros & Cons of Matrix Method



Network Method



Example of Network Method 1/2

Possible Adverse Impacts			Corrective Action	Control Mechanisms
Initial Condition	Consequent Condition	Effects		
Increased surface runoff **(E)**	Flooding **(H)**	Gullying and Erosion **(K)**	Place intermittent flower beds	
Pollution of Ground Water **(F)**	Degradation of Water Supply **(I)**	Health Hazard **(L)**		Building Code
Removal of Topsoil **(G)**	Decreased Fertility **(J)**	Death of Flora **(M)**	Plant Shrubs	Plant Shrubs

Source: Esraz Z., 2017

Action	Impacts			Branches
A	(F)	(I)	(L)	1
B	(E)	(H)	(K)	2
C				
D	(G)	(J)	(M)	3

Example of Network Method 2/2

	Impact Score (1-10 interval Scale)					
Impacts	Magnitude	Importance	Probability of Occurance			
E	5	3	B →	E(0.8)	D →	E(0.7)
F	2	5	A →	F (.5)		
G	3	4	C →	G(.3)	D →	G(.4)
H	4	5	E →	H(.7)		
I	2	9	F →	I(.6)		
J	2	5	G →	J(.8)		
K	3	7	H →	K (.7)		
L	2	10	I →	L (.9)		
M	1	6	J →	M (.8)		

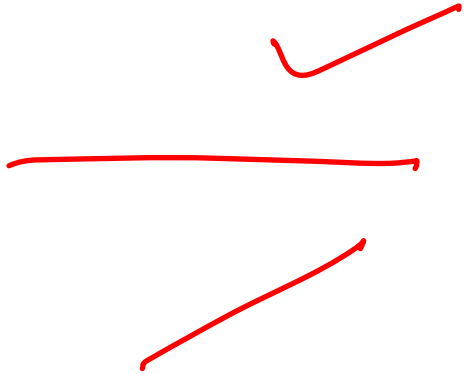
Branch 1 (2)(5)(.5)+(2)(9)(.6)+(2)(10)(.9)=33.8

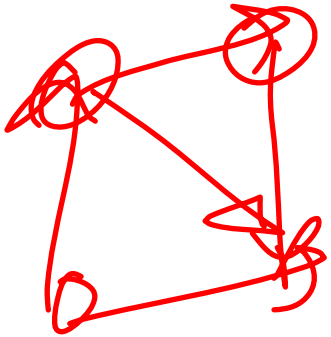
Branch 2 (5)(3)(.8)+(5)(3)(.7)+(4)(5)(.7)+(3)(7)(.7)=51.20

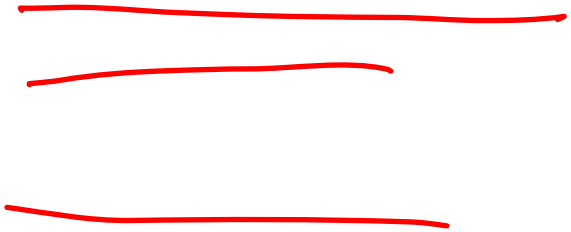
Brach 3 (3)(4)(.3)+(3)(4)(.4)+(2)(5)(.8)+(1)(6)(.8)=21.2

Grand Network Index 33.8+51.2+21.2= 106.2

Pros & Cons of Network Method







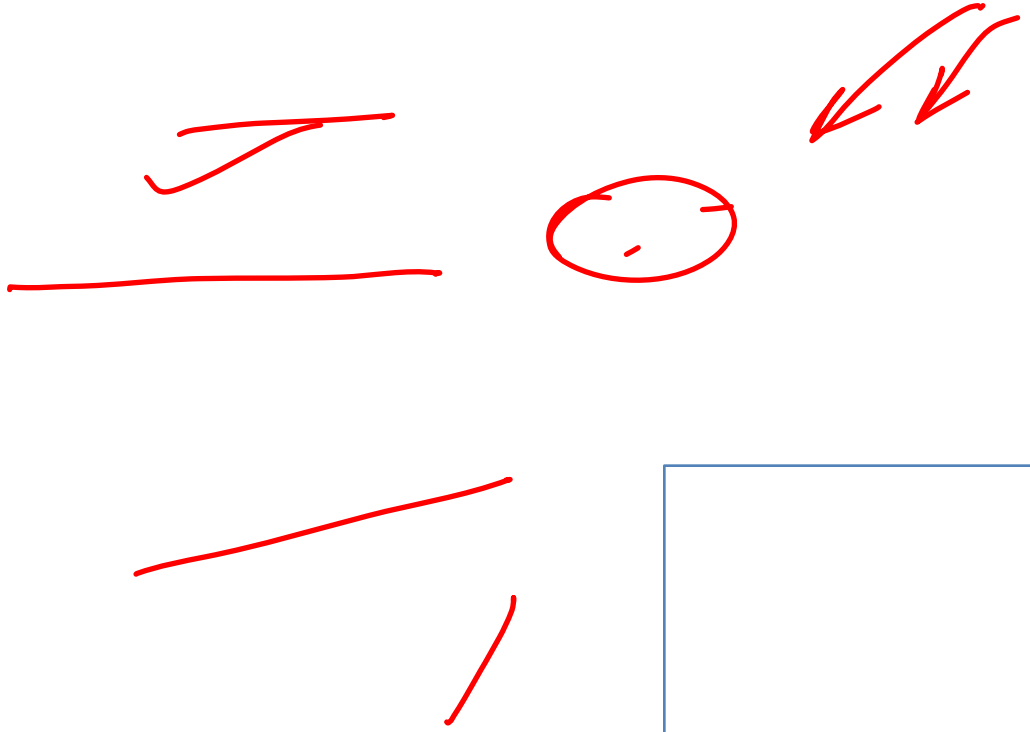


Overlay Method

Set of maps of a project area's environmental characteristics covering physical, social, ecological and aesthetic aspects.

Separate mapping of critical environmental features at the same scale as project's site plan (Ex: wetlands, steep slopes, soils, floodplains, bedrock outcrops, wildlife habitats, vegetative communities, cultural resources, etc.)

Pros & Cons of Overlay Method



Old Vs. New Technique for Overlay Method

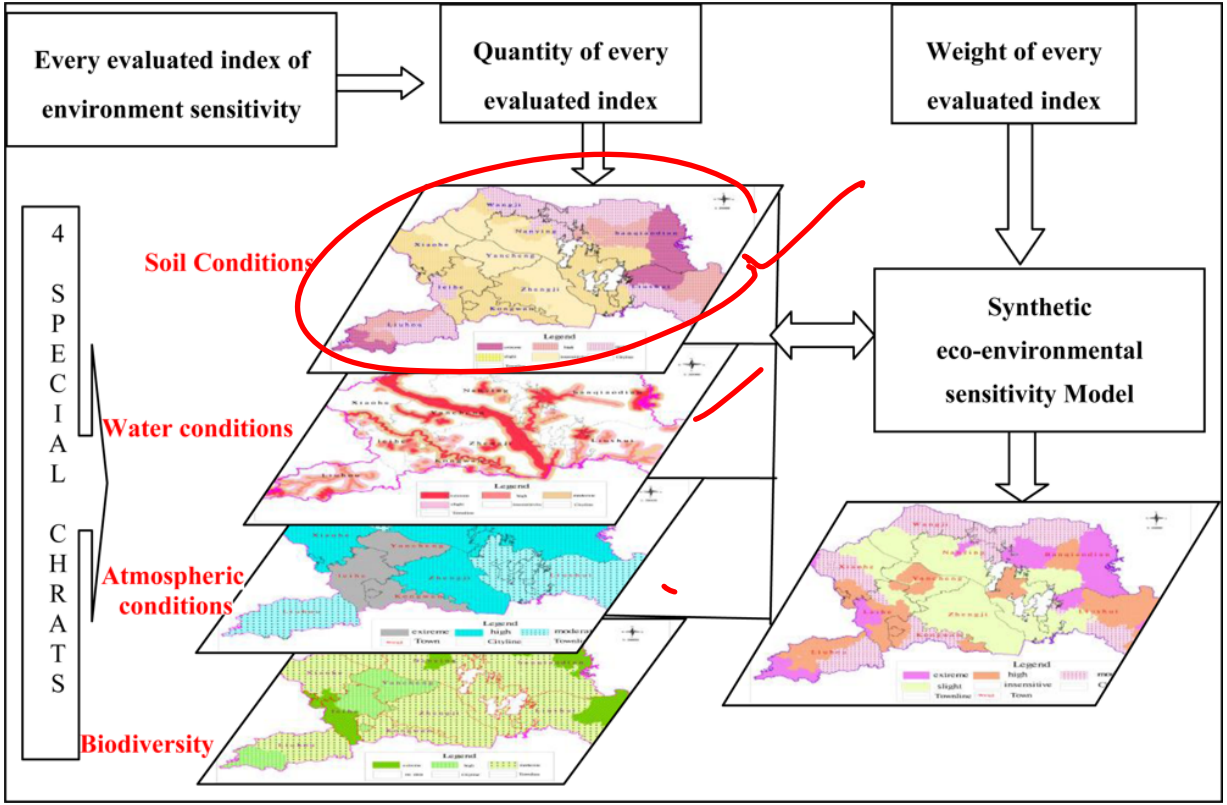
Old technique:

Environmental features were mapped on transparent plastic in different colors.

Modern technique:

It is done using GIS (Geographic Information Systems) with help of skilled personnel.

Example of Overlay Method: GIS technique



Different aspects are formed in layers representing their impact level in Colour coding

Source: Zange J. et al., 2011

Simulation Method

Micro method used for a very specific situation

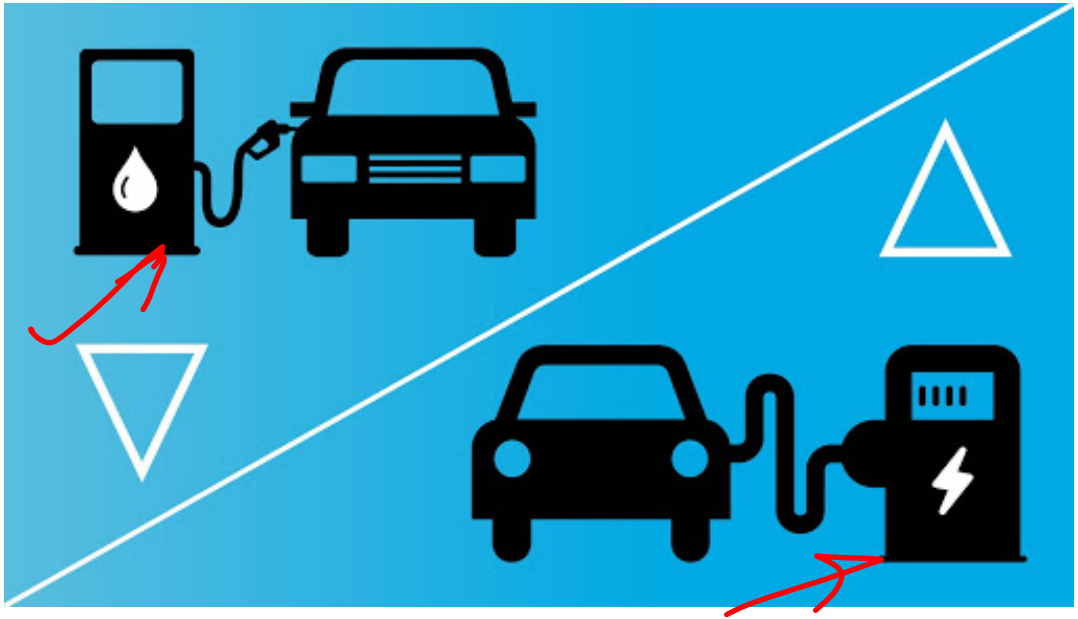
Scope of method is limited to a single activity

Experimental in nature

Assess the impact of any product or small scale service



Example of Electric Vehicles Vs. Fossil Fuel Based Vehicles



- Electric & fossil fuel based vehicles compared in standard conditions for exact emissions and cost per km.
- Simulations are done to compare the environmental impacts by both type of vehicles

Image Source: noodoe.com

Cost-Benefit Analysis (CBA) Method



Social Cost-Benefit Analysis (SCBA)

Assessing the desirability of projects in the public as opposed to the private sector

Identification of costs and benefits (social, environmental & economic)

Generally used by international institutions such as World Bank

Measurement of costs and benefits in monetary terms

The effect of (risk and uncertainty) during project life cycle

Stages of SCBA

Financial profitability of the project based on the market prices

Shadow prices for the resources to arrive at the net benefit of the project at economic process

Adjustment of the net benefit for the projects impact

Adjustment of the net benefit for the projects impact on income distribution

Adjustment of the net benefit for the goods produced whose social values differ from their economic values



Limitations of SCBA

- Measurement of social costs and benefits are formidable.
- Costs and benefits are intangible and their evaluation in terms of money is bound to be subjective.
- Comparison with future projects is difficult.
- Impact on the ecology and people of the particular region and the country as a whole are bound to be differing from case to case.

Environment Index Method

- Standards are predefined.
- Raking is done for various parameters against set standards
- Based on overall score, alternatives are given ranks.

Based on this method, EPI (Environment Performance Index) is developed to rank countries.

Environment Performance Index (EPI) Ranking

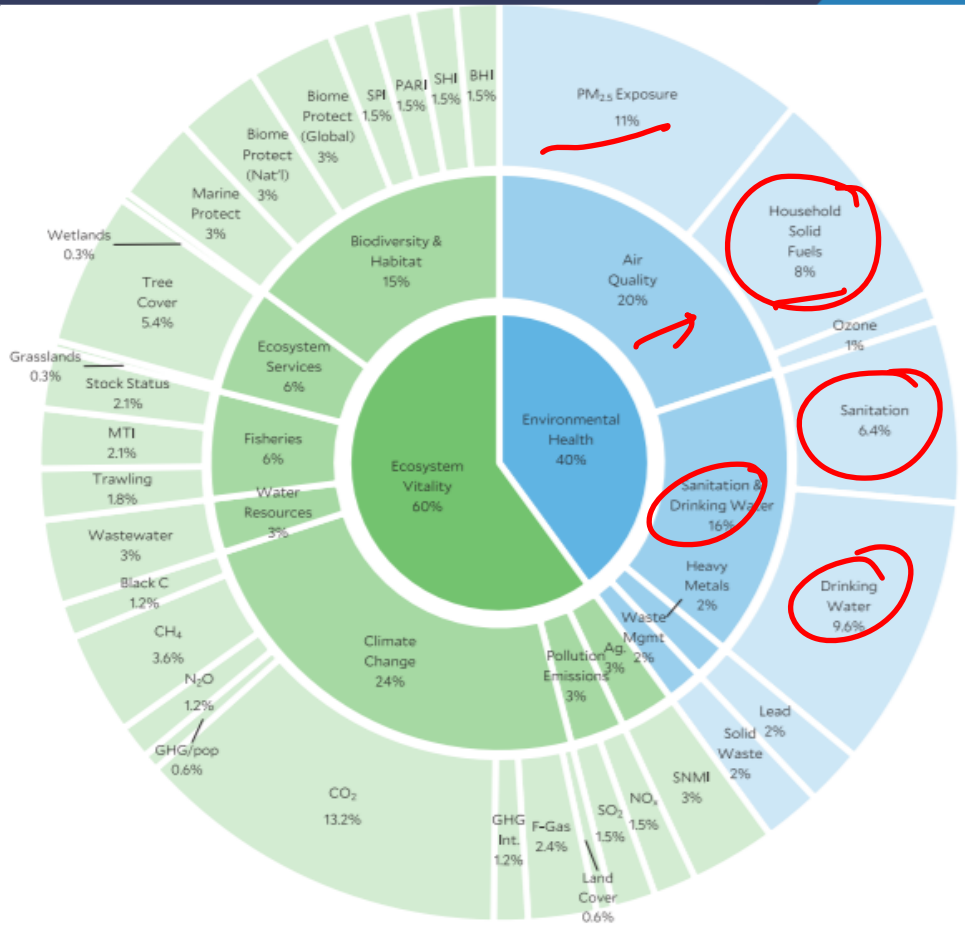
International ranking mechanism for accessing countries' efforts against climate change

The EPI uses the distance-to-target technique for indicator scoring, which situates each country relative to its targets for best and worst performance corresponding to scores of 100 and 0.

32 indicators into 11 issue categories and two policy objectives, with weights shown at each level as a percentage of total score

Source: epi.yale.edu

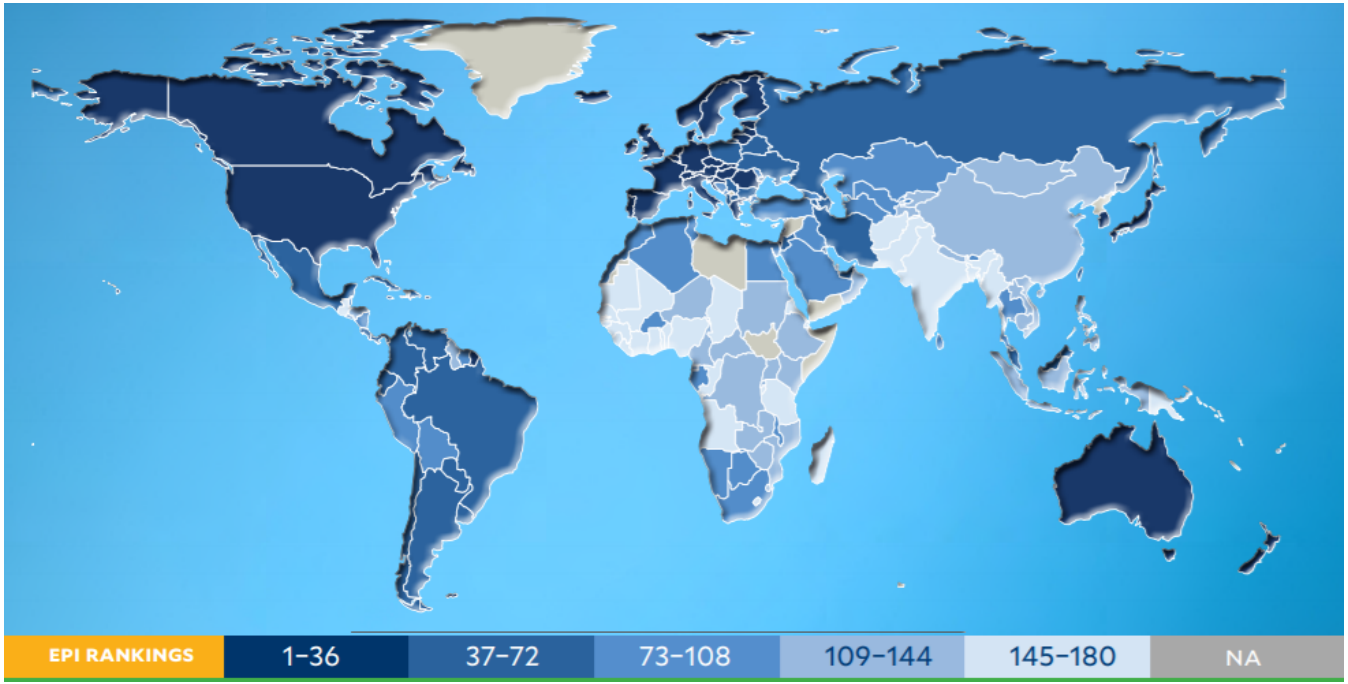
Indicators & Weightage



Source: epi.yale.edu

Image: Weightage given for various indicators

Example of (EPI) Ranking



Source: epi.yale.edu

Environment Performance Index (EPI) Ranking for Countries worldwide

Conclusions

- Various methodologies are available for conducting EIA studies.
- The Methodologies are selected and used as per project requirements and conditions, time constraints, and available data and resources.
- Every methodology has its own benefits and limitations, which can be taken into consideration during its selection.



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Thank You